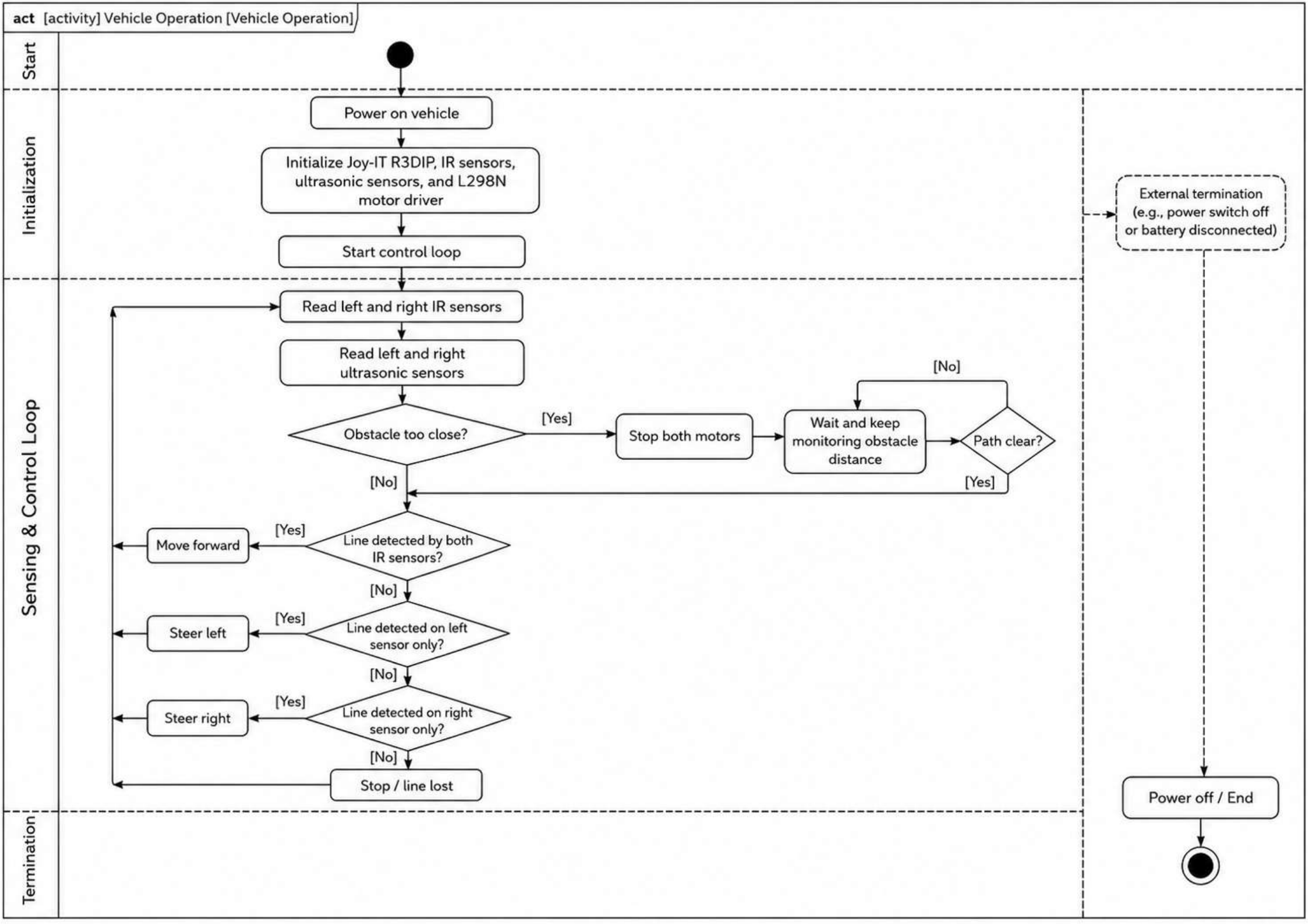

act [activity] Vehicle Operation [Vehicle Operation]
Start
Initialization
Sensing & Control Loop
Termination
Power on vehicle
Initialize Joy-IT R3DIP, IR sensors, ultrasonic sensors, and L298N motor driver
Start control loop
Read left and right IR sensors
Read left and right ultrasonic sensors
Obstacle too close?
[Yes]
Stop both motors
Wait and keep monitoring obstacle distance
Path clear?
[No]
[Yes]
[No]
Line detected by both IR sensors?
[Yes]
Move forward
[No]
Line detected on left sensor only?
[Yes]
Steer left
[No]
Line detected on right sensor only?
[Yes]
Steer right
[No]
Stop / line lost
External termination (e.g., power switch off or battery disconnected)
Power off / End